Promo Finder: finding hook-to-cliffhanger promos in microdrama screenplays

Kuku AI Generalist Assignment, **Problem 1** | Pendem Medhansh | ME23BTECH11046 | Mechanical Engineering | Show: *I Married a Broken Soldier. He was a Duke* (50 episodes)

1. Why this problem, and scope

I picked Problem 1 because the output can be **checked against the source**: every start/end must be a real timestamp, every quoted line must exist, every clip must be 20-60 s. Problem 2 risks invented law and is hard to verify; Problem 3 needs me to prove a problem exists before building. Scope: screenplay text only (no video, audio or engagement data). I find the clips and cut points; no editing, music or overlays.

2. Assumptions (the brief was silent)

- Timestamps `mm:ss.ff` are minutes:seconds with a decimal. A promo = 20-60 s **total** across all its segments.
- Cuts land only on a spoken/action beat boundary and **never inside someone's line** (an editor would never cut mid-sentence).
- Hook = the first 3 s of the clip. Weights: hook 0.35, cliffhanger 0.40, standalone 0.25 (my judgement: the cliffhanger is what earns the next click). Standalone below 2/10 is demoted.
- Stitching (opening hook + a later cliffhanger beat, skipping the middle) is allowed **within one episode only**; cross-episode joins are not built. Global list capped at 2 promos per episode.
- I assumed an episode-ending cut is usually a cliffhanger (true of this show's design). That assumption caused a bad pick (see section 5).

3. How it works: what is code, what is AI, where a human sits

- **Code (no AI): parse + candidates.** Screenplay -> timestamped blocks -> every legal window (16,828 windows scored across 50 episodes in about 1.2 s). Because only code chooses timestamps, a model **cannot invent one**.
- **Code: offline heuristic scorer** (keywords + structure) ranks windows and shortlists 6 clips per episode (188 clips). Free, instant, no key.
- **AI (optional, built, not yet run live): one model call per episode** scores only the shortlist, writes the hook / cliffhanger / one-line reason and flags risks. Quoted lines must appear verbatim in the clip or are replaced. The model is a flag (`--provider gemini|groq|openai|anthropic --model X`); responses are cached.
- **Human:** an editor reviews the ranked top 15 per show and accepts/rejects; their choices become the test set and few-shot examples.

4. Result on the full pack, top 5 (full list: outputs/ranked_promos.csv)

#	Ep / segments	Hook (first 3 s)	Cliffhanger (last line)	H / C / S
1	Ep 15 00:44.8-01:22.4 (37.6s)	Thug With Cap: Get out of the way!	Thug With Cap: What? You want to fight us, veteran?	6.75 / 6.25 / 3.38
2	Ep 2 00:00.0-00:27.4 (27.4s)	The Former Groom: I cannot marry you, Juliana.	Lady Juliana: A better choice? Is that what she is?	6.15 / 4.0 / 6.92
3	Ep 11 01:00.0-01:46.6 (46.6s)	Juliana: Stop! Those are my mother's medicines!	Gerald: There it is. Ha!	8.7 / 4.75 / 3.7
4	Ep 3 01:00.5-01:25.5 (25.0s)	(action) The guards forcefully grab Juliana by both shoulders and drag her back...	(action) The Bearded Stranger effortlessly blocks the guard's strike and drives...	8.4 / 3.75 / 3.54
5	Ep 16 00:42.5-01:23.2 (40.7s)	(action) Alexander lifts the thug onto his tiptoes against the cottage wall, gl...	(action) Alexander and Juliana gaze deeply into each other's eyes, an unspoken...	5.7 / 6.75 / 2.5

Why #1 first: Ep 15 opens on a physical shove and a woman crying over crushed herbs, the hero arrives, and it cuts on a villain's challenge ('You want to fight us, veteran?') so the fight is unresolved. #2 (Ep 2) starts at the episode's first second, which makes it the most standalone clip in the list (7/10): a public rejection that ends on a question. H/C/S = hook / cliffhanger / standalone, 0-10. Scores are harsh by design: the best clip scores 6.2.

5. How I checked quality, and where it picks badly

- **Automated (validate.py, independent of the engine):** every start/end is a real beat boundary, no cut through a spoken line, duration 20-60 s, hook and cliffhanger text found in the right part of the clip. **188/188 shortlisted clips pass** (it caught two real bugs while I built it).
- **Manual:** I read the top 5 in full, #7-#10 as head/tail excerpts, and 3 random low-ranked clips. I did **not** have editor-picked promos, so I cannot report precision; that is the first thing I would add.
- **Fixed after reading:** (a) Ep 4 ranked #2 but ends with the couple walking into the sunlight under a triumphant swell (resolved, not a cliffhanger); cause: episode-end bonus plus 'swell' counted as ominous. (b) A clip started 0.4 s inside a spoken line. (c) A fragment ('They have been--') scored a perfect hook.
- **Still wrong:** (1) Ep 16 (#5) ends on a soft romantic gaze; the real cliffhanger ('Gerald paid us!') is 20 s earlier, and the scorer was lured by the script's own 'cliffhanger hold' sound note. (2) The heuristic reads words, not images: Ep 42's Duke-reveal entrance scores hook 0.0 and sits in the bottom 20%; the Ep 18 fire scene is underrated (lexicon gaps). (3) 'Standalone' is a weak proxy: Ep 11 and Ep 46 lean on earlier plot. (4) #1 and #10 are two halves of one Ep 15 scene; a stitched version may beat both. (5) **The LLM stage has not been run against a live model** (no key); only its plumbing was tested with a mock. Expect it to fix (2)-(3), not guaranteed.

6. Scale to every show

- **Cost per 50-episode show:** heuristic is free (~1 s). LLM stage measured at ~63K input tokens (one call per episode, shortlist only) plus roughly 25-30K output tokens (my estimate). At list prices I remember (verify before relying): about $0.02 on a flash-class model, about $0.6 on a mid-tier one. Weekly new shows are therefore cheap; the cost that matters is editor time, which drops from watching 50 episodes to reviewing 15 clips.
- **Where it breaks at volume:** the keyword lists are tuned to period romance in English. Other genres/languages need different lists, so for them I would widen the shortlist (12+ per episode) or sample windows at a fixed stride, letting the LLM do more of the work. Add: video/audio cues (shot cuts, loudness), cross-episode stitching, and measuring real promo watch-through to re-tune the weights.

7. AI tools used, and one place they got it wrong

Tools: Claude (planning, code, drafting this doc), Python standard library, Streamlit for the viewer. **Wrong:** Claude's first scoring version confidently ranked the Ep 4 happy wedding exit #2 as a 'cliffhanger'. I only caught it by reading the actual clip transcript, not by trusting the score. The fix (warm/resolved cues, no mid-line cuts) is in the code, and the lesson is why the validator and a manual read exist.